Anirban Bairagi

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anirban-bairagi anirbanbairagi

Experience

Doctoral of Philosophy *Paris, France*
Institut d'Astrophysique de Paris, CNRS & Sorbonne Université *Jan 2023 – Feb 2026*

- Developed **PatchNet**, a scalable hierarchical field-level inference framework for next-generation cosmological surveys, combining small-scale patch-based information with large-scale summary statistics. Achieved near-optimal Fisher information recovery relative to full field-level inference while significantly reducing GPU memory usage and simulation cost.
- Derived an empirical **power-law scaling relation** for simulation-based inference like Large Language Models, quantifying the number of simulations required to achieve lossless optimal neural summaries and providing practical guidance for computationally efficient SBI pipelines.
- Developed an analytical displacement-potential method to correct small-scale clustering in fast Particle-Mesh simulations, bridging the gap between fast approximations and full N-body accuracy.
- Leading the **wavelet-based inference analysis** for SDSS data within the **Learning the Universe** collaboration (Simons Foundation). Trained ensembles of **normalizing flows** on wavelet scattering coefficients from fastPM halos and galaxies in redshift space to perform robust neural posterior inference of cosmological parameters.

Technical Consultant *Calcutta, India*
TCG Digital *June 2022 – Dec 2022*

- Automated end-to-end monthly analytics pipeline in **Python**, from **SQL** data extraction to final analysis, reducing manual workload and turnaround time for client reporting.
- Performed **data-driven workforce analytics on leave patterns** in a major U.S. supermarket chain, enabling optimized staffing strategies that reduced revenue loss due to understaffing and overtime costs by 85%.
- Reduced the loss incurred by the pharmaceutical companies by 72% due to insufficient and excessive supply of diagnostic kits in different countries of Europe using **LDA** and **XGBoost**.

Caltech SURF - LIGO *Pasadena, CA*
California Institute of Technology *May 2021 – July 2021*

- **Simulated** laser beam spot images incorporating mirror micro-roughness and CCD sensor noise to mimic real-world optical imperfections in the LIGO detector.
- Developed a **Convolutional Neural Network** (CNN) to infer beam position from noisy CCD images with sub-pixel accuracy ($\leq 40\ \mu m$) to mitigate noise from the detector signal due to misalignment of mirrors.

MITACS Globalink Research Fellow *London, Ontario*
Western University *July 2021 – Oct 2021*

- **Modeled** Continuous Gravitational Wave (CGW) signals from non-precessing triaxial neutron stars.
- Built a real-time CGW detection algorithm using **Convolutional Neural Network** (CNN), enabling prompt identification of potential electromagnetic counterparts.
- Performed Bayesian inference on the signals using **Markov Chain Monte** Carlo (MCMC) to obtain robust posterior distributions for astrophysical parameters.

Education

Institut d'Astrophysique de Paris, CNRS & Sorbonne Université *Jan 2023 – Feb 2026*
PhD is Astrophysics, Statistics and Machine Learning ***Simons Foundation***

- **Advisor: Dr. Benjamin Wandelt**

Indian Institute of Technology Kharagpur *July 2017 – April 2022*
B.Sc.-M.Sc. in Physics ***CGPA: 8.55/10***

- **Coursework:** General Relativity, Astrophysics, Mathematics, Statistics, Deep Learning

Technical Skills

Languages: Python, Cython, Mathematica, MATLAB, C, SQL, HTML, CSS, Arduino
Machine Learning: XGBoost, Deep Learning, CNN, YOLO, Diffusion models, Normalizing flows, Transformers
Frameworks/Libraries: Numpy, Pandas, Scipy, Scikit-learn, Matplotlib, Pytorch, TensorFlow, Keras, OpenCV
Tools: Weights & Biases, Git, Linux, CMake

Publications

1. **A. Bairagi**, B. Wandelt, F. Villaescusa-Navarro, *The BIG SOBOL SEQUENCE: How many simulations do we need for simulation-based inference in cosmology?*, *A&A* 703 (2025) A301, arXiv:2503.13755.

2. **A. Bairagi**, B. Wandelt *PatchNet: A hierarchical approach for neural field-level inference from Quijote Simulations*, arXiv:2509.03165, accepted at *JCAP*.

3. M. Cagliari, **A. Bairagi**, B. Wandelt, *Hierarchical summaries for primordial non-Gaussianities*, arXiv:2510.12715, submitted to *A&A*.

4. **A. Bairagi**, N. Chartier, D. Bartlett, B. Wandelt, *Point-Set Clustering Correction Using a Displacement Potential*, manuscript in preparation.

5. **A. Bairagi**, Y. Drori, T. Edo, R. Adhikari, *LIGO Laser Beam Tracking*, LIGO Technical Document LIGO-T2100205, 2021. https://dcc.ligo.org/LIGO-T2100205/public

6. **A. Bairagi**, *Gravitational Wave Detection and Glitch Classification Using Convolutional Neural Networks*, Poster presented at the Royal Astronomical Society. https://ras.ac.uk/poster-contest/anirban-bairagi